

Risk-Based ETF Portfolio Construction

Selection Methodology, Investor Profiles & ETF Specifications

A personal investment track-record project demonstrating asset allocation
and risk-management skill across three risk profiles.

September 2026

Executive Summary

This document sets out the construction of three ETF-based portfolios — Risk-Lover, Risk-Neutral, and Risk-Averse — built from a shared universe of eight exchange-traded funds. Each portfolio reflects a distinct, internally consistent risk profile, dialled primarily through two levers: the size of the fixed-income (bond) allocation, and the presence or absence of higher-volatility satellite positions (small-cap equity, emerging markets, and bitcoin). The purpose is to demonstrate a genuine, evidence-based asset-allocation process — not to pick individual winning stocks — and to document that process clearly enough that a reader can verify the reasoning behind every allocation choice.

Each portfolio is paired with a live tracking workbook that logs trades, prices, and performance against a pure S&P; 500 benchmark, so the process can be evaluated on real, forward-dated results over time rather than a backtested hypothesis alone.

What this document covers

- The three investor profiles this project is designed around, and why each allocation fits its profile.
- The construction logic connecting all three portfolios — what changes, what stays constant, and why.
- A full specification sheet for every ETF used, so the underlying holdings are transparent and verifiable.

Investor Profiles

Each portfolio targets a distinct investor archetype. The profiles below describe the characteristics each portfolio is designed to fit — not a single real individual, but a defensible, realistic persona used to justify each allocation decision.

Risk-Lover

- **Horizon:** Long (25–30+ years) — ample time to recover from deep drawdowns.
- **Objective:** Maximise long-run growth; short-term volatility is an accepted cost, not a concern.
- **Behavioural fit:** Comfortable holding a position through a 30–50%+ decline without abandoning the thesis.
- **Knowledge level:** Understands factor investing and crypto-asset risk well enough to size a satellite position deliberately rather than chase headlines.
- **What this rules in:** Concentrated factor tilts (small-cap, EM), a meaningful bitcoin allocation, zero bonds.

Risk-Neutral

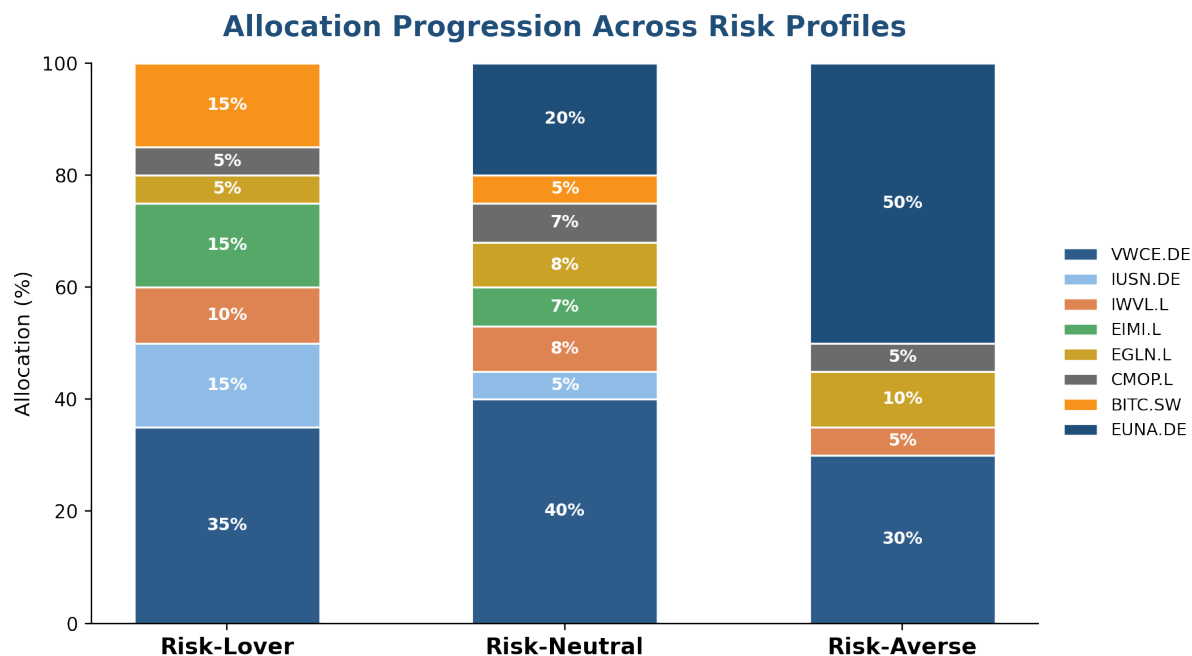
- **Horizon:** Long-to-medium (15–30 years) — still growth-oriented, but wants a smoother ride.
- **Objective:** Balance growth with real downside protection; avoid the largest single-asset drawdowns.
- **Behavioural fit:** Comfortable with volatility in principle, but benefits from an allocation that actively dampens it rather than relying purely on time horizon.
- **Knowledge level:** Understands diversification benefits and is willing to hold a meaningful bond position even though it caps some upside.
- **What this rules in:** A real (20%) bond allocation, moderate factor tilts, a small speculative satellite.

Risk-Averse

- **Horizon:** Still long by conventional standards, but priorities skew toward capital preservation over maximum growth — low tolerance for large drawdowns regardless of time to recover.
- **Objective:** Protect capital while still participating in long-run equity growth at a reduced scale.
- **Behavioural fit:** Would likely act on a large drawdown (sell, panic, deviate from plan) if not structurally protected by a large ballast allocation.
- **Knowledge level:** Prioritises simplicity and lower variance in outcomes over the possibility of higher but less certain returns.
- **What this rules in:** A dominant (50%) bond allocation, no speculative satellites, no small-cap or EM overweight — only a single, historically lower-beta equity factor (value) retained alongside the core.

Portfolio Construction Methodology

All three portfolios are built from the same eight-ETF universe, weighted differently. This is a deliberate design choice: rather than choosing different holdings for each risk profile, the same building blocks are dialled up or down, which makes the reasoning behind each portfolio easier to verify and compare.



Allocation progression across the three risk profiles, by ETF.

The four levers used to dial risk

Fixed income (EUNA.DE) is the primary risk dial.

0% → 20% → 50% as risk tolerance falls. This is the single biggest lever, consistent with standard portfolio theory — the equity/bond mix drives most of a portfolio's overall risk profile.

Speculative satellites are the first thing dropped.

Bitcoin goes 15% → 5% → 0%, and small-cap + EM (the highest-volatility equity factors) go from a combined 30% down to 0% for the risk-averse investor. These add volatility without a corresponding ballast, so they are cut before anything else.

Core cap-weighted equity never disappears.

35% → 40% → 30% across the three profiles. Even the risk-averse version keeps a meaningful equity core — defensible for an investor with a genuine long-term horizon rather than a near-term liquidity need.

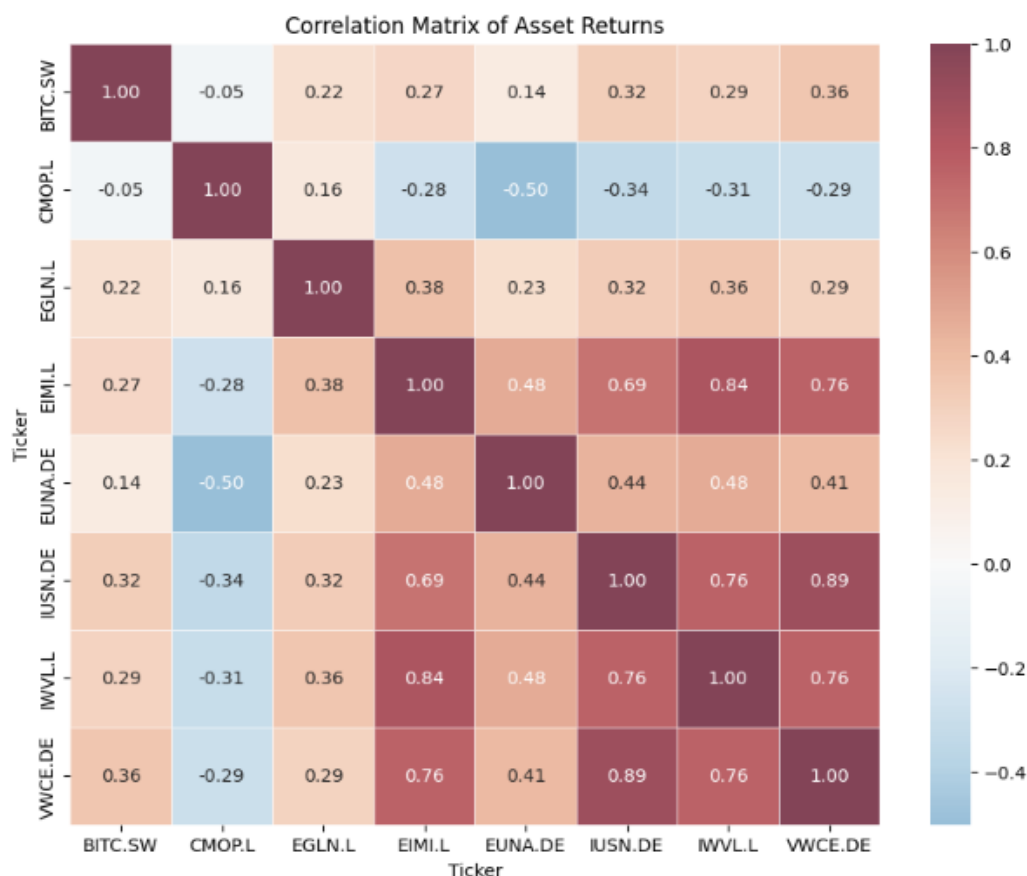
The real-asset hedge stays roughly stable.

Gold + commodities move only slightly (10% → 15% → 15%) across all three profiles. This is intentional: their diversification benefit doesn't depend much on risk appetite, so it's the one sleeve that barely moves.

This progression is also supported by prior correlation analysis on this same ETF universe: gold and commodities showed negative-to-low correlation with the equity sleeve (roughly -0.34 to +0.38), while the equity factor tilts (small-cap, value, EM) were highly correlated with the core (0.68–0.89) — meaning those tilts add conviction to the equity view rather than true diversification, which is exactly why they are the first exposures reduced as risk tolerance falls.

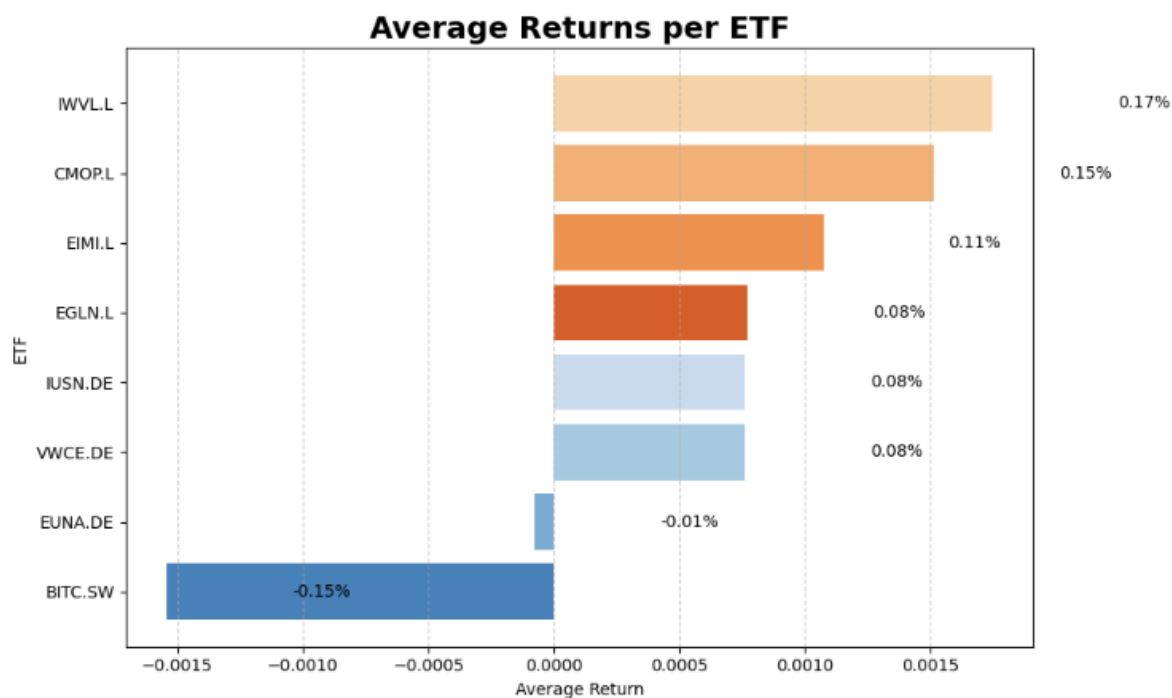
Backtest Analysis: Trailing 12-Month Data

The two charts below were generated in Python using each ETF's trailing 12-month price history (from 13 September 2025 to 13 September 2026, the date of writing). They depict the empirical relationships this methodology relies on — correlation between the eight ETFs, and each fund's average daily return over the period — as observed as of today. This is a retrospective snapshot, not a forecast: both figures will shift as the trailing window rolls forward, and should be regenerated periodically rather than treated as fixed inputs.



Reading this matrix

VWCE, IUSN, IWVL, and EIMI move together closely (0.69–0.89) — confirming that these four equity holdings are concentrating equity-risk conviction rather than diversifying against each other. EUNA.DE (bonds) shows the most consistently low-to-negative correlation with the equity block (0.41–0.48 with equities, but -0.50 with CMOP.L), making it the strongest true diversifier in the set — the empirical basis for using it as the primary risk dial across the three portfolios. CMOP.L (commodities) is negatively correlated with nearly everything else, including bonds, reinforcing its role as a diversifier rather than a directional bet. BITC.SW remains only mildly correlated with the rest (0.14–0.36) — lower than equities, but not the near-zero relationship sometimes claimed for bitcoin as a diversifier.



Average daily return per ETF, trailing 12 months as of 13 September 2026.

Reading this chart

IWVL (value) and CMOP (commodities) led on average daily return over this window, ahead of the VWCE core — a data point that supports, though does not prove, the factor-tilt thesis behind the Risk-Lover and Risk-Neutral portfolios. EUNA.DE (bonds) shows a small negative average return over this specific 12-month window, which is a reasonable outcome for a low-volatility asset during a period of rate volatility, and is not inconsistent with its role as a portfolio stabiliser rather than a return driver. BITC.SW was the only clearly negative-average-return position in this window — worth stating plainly rather than omitting, especially since it is the highest-conviction, highest-risk position in the Risk-Lover portfolio. Past performance over a single 12-month window is not predictive of future results.

Risk-Adjusted Performance (Sharpe & VaR)

Beyond raw returns and correlation, two standard risk metrics were calculated for each of the three full portfolios (not individual ETFs) over the same trailing 12-month window, as of 13 September 2026: the annualised Sharpe ratio and the 95% one-day historical Value at Risk (VaR).

Portfolio	Sharpe Ratio	95% Daily VaR
Risk-Lover	0.40	-1.65%
Risk-Neutral	-0.06	-1.87%
Risk-Averse	0.02	-1.59%

Calculated over the trailing 12 months as of 13 September 2026; risk-free rate assumption $R_f = 2.487\%$ (annualised).

Note on the risk-free rate used. $R_f = 0.02487$ (2.487%) is the Germany 3-Month Bond Yield at the time of writing. This proxy was chosen because: (1) all three portfolios are EUR-denominated, so the risk-free comparison should be a euro-area rate, not a USD rate such as U.S. Treasury bills; (2) German Bunds are the conventional euro-area benchmark for a “risk-free” sovereign rate, carrying the lowest credit risk within the eurozone; and (3) a short (3-month) maturity was used specifically to avoid introducing interest-rate/duration risk into the risk-free reference itself — a longer-dated bond yield would itself carry price risk, which defeats the purpose of a “risk-free” baseline. This is a standard, defensible convention, but it is a choice: using a different maturity, currency, or issuer would shift every Sharpe ratio in the table above by a small, uniform amount.

What these metrics mean

Sharpe Ratio

Measures return earned per unit of risk taken: $(\text{portfolio return} - \text{risk-free rate}) \div \text{portfolio volatility}$, annualised. A Sharpe ratio above 0 means the portfolio outperformed a risk-free asset (here, a 2.487% annual rate — see justification below) after adjusting for how much volatility it took on; a Sharpe ratio below 0 means the portfolio's return over the period didn't even clear the risk-free rate, regardless of how much risk was taken. Higher is better, and values are only meaningfully comparable across portfolios measured over the same window.

95% Daily Value at Risk (VaR)

An estimate, based on this portfolio's actual historical daily returns, of the loss threshold that should not be exceeded on 95% of trading days. A VaR of -1.65% means that, historically, this portfolio lost more than 1.65% in a single day on roughly 5% of trading days (about one day per month) — the other 95% of days saw a smaller loss or a gain. A larger-magnitude (more negative) VaR indicates a portfolio with fatter downside tails on an average day, not necessarily a worse portfolio overall.

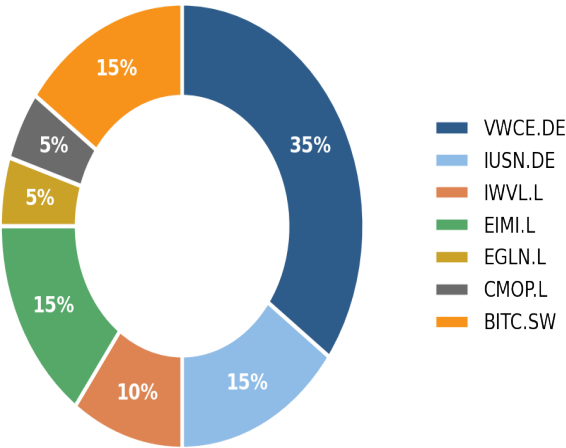
Reading the results

Risk-Lover recorded the highest Sharpe ratio (0.40) despite having zero bonds and the largest satellite exposure — in this specific window, its factor tilts and equity conviction were rewarded enough to clear the risk-free rate comfortably per unit of volatility taken. Risk-Averse posted a small positive Sharpe ratio (0.02) and, as expected by design, the smallest-magnitude VaR (-1.59%) — the lowest day-to-day tail risk of the three, consistent with its 50% bond ballast.

Note on the Risk-Neutral result. Despite being the intentionally 'middle' profile by design, Risk-Neutral recorded both the lowest Sharpe ratio (-0.06 — the only negative value of the three) and the largest-magnitude VaR (-1.87%) over this specific 12-month window. This is a sample-specific outcome of the realised price path, not evidence that a moderate-risk design is structurally worse than an aggressive or conservative one. Two contributing factors: its modest satellite positions (small-cap, EM, bitcoin) added volatility without this particular period's returns being large enough to compensate the way they did in the larger Risk-Lover allocation, while its 20% EUNA.DE bond position posted a mildly negative average return over the same window (see previous chart), providing less offsetting ballast than intended. A single 12-month sample is a small, noisy dataset for Sharpe ratio and VaR estimation — both statistics can shift materially as more live data accumulates, and should not be treated as a verdict on the Risk-Neutral portfolio's design until a longer track record is available.

Risk-Lover Portfolio

Risk-Lover Portfolio

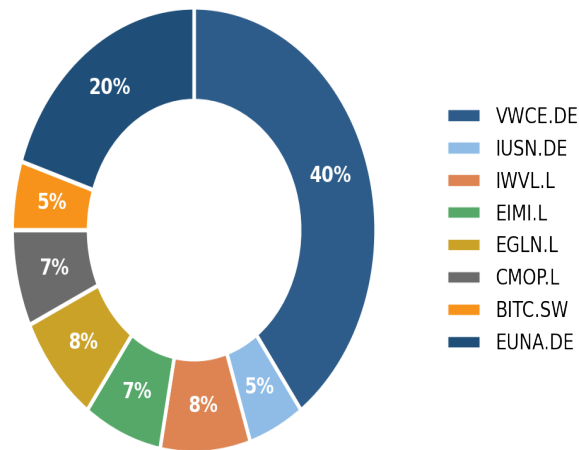


Ticker	Role	Weight
VWCE.DE	Core global equity	35%
IUSN.DE	Small-cap growth tilt	15%
IWVL.L	Value factor tilt	10%
EIMI.L	Emerging markets growth tilt	15%
EGLN.L	Real-asset diversifier	5%
CMOP.L	Real-asset diversifier	5%
BITC.SW	High-conviction satellite	15%

Zero bonds, the largest combined equity-tilt exposure (40% across small-cap, value and EM), and the largest bitcoin allocation of the three profiles (15%). This portfolio accepts the highest expected volatility in exchange for the highest expected long-run return, and is appropriate only for an investor who has both the time horizon and the temperament to hold through severe drawdowns without deviating from the plan.

Risk-Neutral Portfolio

Risk-Neutral Portfolio

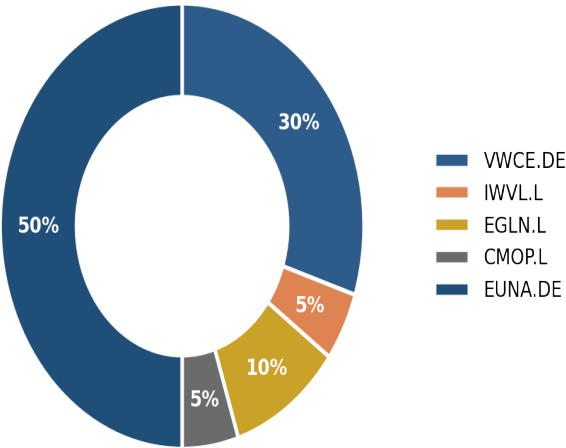


Ticker	Role	Weight
VWCE.DE	Core global equity	40%
IUSN.DE	Small-cap growth tilt	5%
IWVL.L	Value factor tilt	8%
EIMI.L	Emerging markets tilt	7%
EGLN.L	Real-asset diversifier	8%
CMOP.L	Real-asset diversifier	7%
BITC.SW	Speculative satellite	5%
EUNA.DE	Fixed income ballast	20%

A genuinely balanced allocation: a real 20% bond position provides ballast, factor tilts are present but scaled back from the risk-lover profile, and the bitcoin satellite is small enough to add optionality without materially raising portfolio-level risk. This sits deliberately between the other two profiles on every dimension.

Risk-Averse Portfolio

Risk-Averse Portfolio



Ticker	Role	Weight
VWCE.DE	Core global equity	30%
IWVL.L	Value factor tilt (defensive)	5%
EGLN.L	Real-asset / inflation hedge	10%
CMOP.L	Real-asset diversifier	5%
EUNA.DE	Fixed income ballast	50%

Speculative satellites (bitcoin) and higher-volatility equity tilts (small-cap, EM) are dropped entirely. Bonds dominate at 50%, doing most of the work of dampening volatility. The single equity tilt retained — value — is chosen because it is historically lower-beta than growth or small-cap, making it a quality tilt rather than a risk-seeking one.

ETF Specification Sheet

Full specifications for every ETF used across the three portfolios, so the underlying holding is fully transparent — what it tracks, what it costs, where it's domiciled, and how it's structured. All funds listed are accumulating (dividends/interest are reinvested rather than paid out), per the project's design requirement.

VWCE.DE Vanguard FTSE All-World UCITS ETF (USD) Acc	
ISIN	IE00BK5BQT80
Asset class	Equity — Global, all-cap
Index tracked	FTSE All-World
TER (annual cost)	0.14%
Domicile	Ireland
Replication method	Physical (sampling)
Distribution policy	Accumulating
Role in this project	Core global equity holding across all three portfolios.

IUSN.DE iShares MSCI World Small Cap UCITS ETF Acc	
ISIN	IE00BF4RFH31
Asset class	Equity — Developed markets, small-cap
Index tracked	MSCI World Small Cap
TER (annual cost)	0.35%
Domicile	Ireland
Replication method	Physical (sampling)
Distribution policy	Accumulating
Role in this project	Small-cap growth tilt (Risk-Lover, Risk-Neutral only).

IWVL.L iShares Edge MSCI World Value Factor UCITS ETF Acc	
ISIN	IE00BP3QZB59
Asset class	Equity — Developed markets, value factor
Index tracked	MSCI World Enhanced Value
TER (annual cost)	0.30%
Domicile	Ireland
Replication method	Physical (full replication)
Distribution policy	Accumulating
Role in this project	Value factor tilt — the only factor tilt retained in all three portfolios, including Risk-Averse.

EIMI.L iShares Core MSCI EM IMI UCITS ETF Acc	
ISIN	IE00BKM4GZ66
Asset class	Equity — Emerging markets, all-cap
Index tracked	MSCI Emerging Markets IMI
TER (annual cost)	0.18%
Domicile	Ireland
Replication method	Physical (full replication)
Distribution policy	Accumulating
Role in this project	Emerging-markets growth tilt (Risk-Lover, Risk-Neutral only).

EGLN.L iShares Physical Gold ETC	
ISIN	IE00B4ND3602
Asset class	Commodity — Precious metal (physical)
Index tracked	Spot gold price
TER (annual cost)	0.12%
Domicile	Ireland
Replication method	Physically backed (allocated LBMA Good Delivery gold bars)
Distribution policy	N/A — no income by nature
Role in this project	Real-asset / inflation hedge, present in all three portfolios.

CMOP.L Invesco Bloomberg Commodity UCITS ETF Acc	
ISIN	IE00BD6FTQ80
Asset class	Commodities — Broad basket (energy, metals, agriculture)
Index tracked	Bloomberg Commodity Index
TER (annual cost)	0.19%
Domicile	Ireland
Replication method	Synthetic (swap-based, futures exposure)
Distribution policy	Accumulating
Role in this project	Broad commodity diversifier, present in all three portfolios.

BITC.SW CoinShares Physical Bitcoin ETP	
ISIN	GB00BLD4ZL17
Asset class	Crypto-asset (single, physical)
Index tracked	Spot bitcoin price
TER (annual cost)	0.15%
Domicile	Jersey
Replication method	Physically backed (fully collateralised by held bitcoin)
Distribution policy	N/A — no income by nature
Role in this project	High-conviction speculative satellite (Risk-Lover, Risk-Neutral only; excluded from Risk-Averse).

EUNA.DE iShares Core Global Aggregate Bond UCITS ETF EUR Hedged Acc	
ISIN	IE00BDBRDM35
Asset class	Fixed income — Global investment-grade bonds, EUR-hedged
Index tracked	Bloomberg Global Aggregate Bond (EUR Hedged)
TER (annual cost)	0.10%
Domicile	Ireland
Replication method	Physical (sampling)
Distribution policy	Accumulating
Role in this project	Fixed-income ballast — absent from Risk-Lover, the dominant position (50%) in Risk-Averse.

Updating This Document With Live Results

This document currently presents a construction methodology and a trailing 12-month backtest — it does not yet contain any live, forward-dated performance. Once the tracking period has run for a meaningful length of time (a minimum of 6–12 months is recommended before drawing conclusions), this section is designed to be replaced with actual results.

What to provide when that time comes

- Final Dashboard figures from each tracker workbook (Total Return, CAGR, Max Drawdown, Volatility, Alpha vs. the S&P; 500 benchmark) for each of the three portfolios.
- Any rebalancing events that occurred — dates, trades, and the reasoning logged in each Trade Log — so the write-up can describe process discipline, not just outcomes.
- The actual time period covered (start date to results date), so the report accurately states how long the live track record runs.
- Anything that didn't go to plan — a missed rebalance, a position that needed unplanned adjustment — since a credible track record includes friction, not just clean results.

With that information, this section can be rewritten to present the live results alongside the backtest — clearly distinguishing “what the historical data suggested” from “what actually happened,” which is precisely the comparison that gives a track record its credibility.

Important Notes

This document is a personal investment track-record project intended to demonstrate asset-allocation reasoning and process discipline. It is not financial advice, a recommendation to buy or sell any security, or an offer to manage third-party assets. All example prices shown in the accompanying tracker workbooks are illustrative snapshots taken at the time of writing and should be replaced with actual purchase and check-in prices as trades occur. Past performance of any asset class or ETF does not guarantee future results. Correlation figures referenced in the methodology section are drawn from a trailing 12-month backtest and are not predictive of future correlation behaviour, which can shift materially in stressed markets. Fees, spreads, and taxes are not modelled in the tracker workbooks and will affect real-world returns. Anyone considering managing money on behalf of others, even informally, should seek independent legal and regulatory advice before doing so.